

- The pen is mounted on a moveable arm.
- Colour drawing can be produced by some plotters through interchangeable pens.
- Able to draw output in small size as A4 size or can able to generate very large size can be up to 20ft by 50ft.
- Specially used in the design of cars, ships, aircrafts, buildings, highways etc.

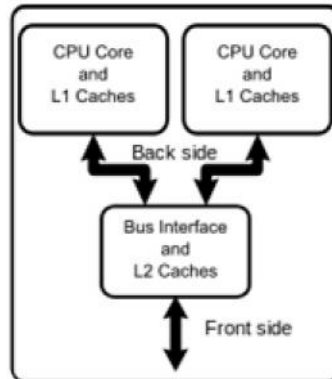
9. WRITE A BRIEF NOTE ON SPEAKER.

- The speaker is output device which is connected to computer's soundcard.
- The speaker output the sound generated by the sound card.
- Audio data is generated by computer is send to audio card which is located inside extension slot.
- It can translate data into audio signal which are sending to speaker.
- The speaker can able to produced series of different tones.

UNIT: 4 INTERNAL/EXTERNAL PARTS WITH COMPUTER CABINATE

1. EXPLAIN TYPES OF PROCESSORS.

DUAL CORE



- Dual-core refers to a CPU that includes two complete execution cores per physical processor.
- It has combined two processors and their caches and cache controllers onto a single integrated circuit (silicon chip)
- Multi-core is similar to dual-core in that it is an expansion to the dual-core technology which allows for more than two separate processors.
- Dual-core refers to a CPU that includes two complete execution cores per physical processor.
- It combines two processors and their caches and cache controllers onto a single integrated circuit (silicon chip).
- It is basically two processors, in most cases, residing side-by-side on the same die.
- Dual-core processors are well-suited for multitasking environments because there are two complete execution cores instead of one.
- Each with an independent interface to the front side bus.
- Since each core has its own cache, the operating system has sufficient resources to handle most compute intensive tasks in parallel.

ADVANTAGES:

- Performance is faster than single-core processors.
- Able to divide information for processing by multiple units.
- Core processor uses slightly less power than two coupled single-core processors

- Multi-core chips also allow higher performance at lower energy.

CORE 2 DUO

- Core 2 Duo is the name given by Intel to its second batch of dual core processors.
- Desktop PCs with the Intel® Core™2 processor family deliver faster performance, greater energy efficiency, and more responsive multitasking.
- Intel's dual core processors were simply 2 Pentium 3 processors that were fabricated in a single chip.
- As they refined their product more, they decided to differentiate their second set of processors from the Core Duo and decided to call it Core 2 Duo.
- The difference between dual core processors and the Core 2 Duo processors is just in the semantics as Core 2 Duo is simply a name given to a more recent family of dual core processors.
- If we translate this to the single core processors, we can say that Core Duo is Pentium 1 while Core 2 Duo is Pentium 2. But all these are still single core processors.
- We can therefore say that Core 2 Duo is simply a subset of all the dual core processors that are out in the market today.

Features and benefits

- With an Intel® Core™2 Duo processor you will get performance-rich technologies,
- Intel® multi-core processing provides greater multitasking performance by combining two independent processor cores in one physical package.
- Execution improves execution time and energy efficiency with more instructions per clock cycle.
- Power Capability enables smarter, more energy-efficient performance.
- Intel® Smart Memory Access improves system performance by optimizing the use of the available data bandwidth
- Intel® Advanced Smart Cache enables higher performance and more efficient cache subsystem by optimizing for multi-core processors.

2. EXPLAIN PRIMORY STORAGE DEVICE.

- It's a temporary storage.
- It consists of some chips.
- The data & instruction are resided in this memory when the CPU executing programs.
- This memory can capable to store & retrieved data very quickly.
- Primary memory is only the memory that is directly access to the CPU.

RAM

- The complete name of RAM is random access memory which is also known as Primary memory.
- It is called read/write memory because data can be read as well as write in RAM.
- It is called random access because you can directly access any data from RAM if you know row & column cell.
- The RAM chip is fixed on the mother board & the mother board is designed in such a way that its memory capacity can be enhanced by adding more RAM chip.
- RAM is a VOLETILE memory.
- RAM chips are of two types:

DRAM:

- Dynamic Random Access Memory is a volatile memory that allows fast access to data and is ideal for use as the primary store of computer systems.
- However, the information is stored as electrical charges and the charges need to be constantly refreshed in order for the data to be maintained.

SRAM:

- Static Random Access Memory is also a volatile memory.
- Once data is written into the chip, it is maintained as long as power is supplied to it; it does not need refreshing.
- However, SRAM is slower than DRAM and it is also more expensive.

ROM

- The complete name of ROM is read only memory.
- The data stored permanently & can't be altered by the programmer.
- Data stored in RAM chip can be read & used but cannot be changed.
- This memory also known a field storage permanent storage or dead storage.
- It is basically used to store manufacturer programmed & user program.
- Most of the basic operations are carried out by electronic circuits which are known as micro programs.
- These programs are stored in ROM. For ex. System Boot Loader.

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PROM

- Programmable Read Only Memory is a non-volatile memory which allows the user to program the chip with a PROM writer.

- The chip can be programmed once, thereafter, it cannot be altered.

EPROM & EEPROM

- Erasable Programmable Read Only Memory and Electrically Erasable Programmable Read Only Memory chips can be electrically programmed.
- Unlike ROM and PROM chips, EPROM chips can be erased and reprogrammed.

3. EXPLAIN FOLLOING PORTS

SERIAL PORT

- Known as asynchronous port or RS-232-C port.
- This type of port is sends & receives data using only two line.
- Therefore this type of port is ideal for connection to the phone circuits which uses 2 data lines.
- The communication process of data transfer is slower.
- Due to this reason it is not used for printer.
- It has high signal travelling capacity.

PARALLEL PORT

- Known as centronics or printer port.
- It's a type of socket found on personal computer for connecting various types of computer devices.
- Normally it is reserved for printer.
- They carry 8 bits at the same time so that communication becomes very faster.
- Due to the faster data communication capability it is used in input & output devices.
- The signal travel capacity is less than serial port.

USB PORT

- It's a new interface technology which are used to connect computer peripherals such as keyboard, mouse, scanner, joysticks, printer, web camera etc.
- USB operate at two speeds 1.5 mbps & 12 mbps.
- The speed is depending upon the devices which are attached with the port.
- For example the devices such as mouse, keyboard Use the low band while digital camera use high speed channel/
- Main advantage is that when devices are attached the appropriate drivers are loaded automatically.

PS/2 PORT

- Developed by IBM for connecting mouse or keyboard to PC.
- It supports the plug that contains 6 pins.
- Also called mouse port.

4. EXPLAIN FOLLOWING CABLES

Serial Cable

- Serial Cables are typically used for RS-232 communication.
- A serial cable is a cable that can be used to transfer information between two devices using serial communication.
- The form of connectors depends on the particular serial port used.
- The maximum working length of a cable varies depending on the characteristics of the transmitters and receivers.
- This cable has short transmission distance because of noise limiting the transmission of high numbers of bits per second when the cable is more than 15 meters long.
- It is cheap to purchase and is simple to join and connect.
- It is suitable for unbalanced data standards.
- Only one device can be connected to the cable.

Parallel Cable

- On many legacy peripherals, the parallel cable utilized both the 25 pin Sub-D connector and the 36 pin Centronics connector.
- This was a common printer interface and is still in service in great numbers.
- With the advent of "intelligent" laser and ink jet printers, the IEEE-1284 bi-directional printer cable was introduced.
- This parallel interface allows for bi-directional communication resulting in speeds up to 10 times faster than conventional cables.

USB Cable

- To Connecting a USB device to a computer is simple -- you find the USB connector on the back of your machine and plug the USB connector into it.
- If it's a new device, the operating system auto-detects it and asks for the driver disk.
- If the device has already been installed, the computer activates it and starts talking to it.
- USB devices can be connected and disconnected at any time.

5. SHORT NOTE: GRAPHIC CARDS.

- A graphics card is the component in your computer that handles generating the signals that are sent to the monitor or "graphics".
- It is responsible for generating all the text and pictures that are displayed on your screen.

- The images you see on your monitor are made of tiny dots called pixels.
- At most common resolution settings, a screen displays over a million pixels, and the computer has to decide what to do with everyone in order to create an image.
- To do this, it needs a translator -- something to take binary data from the CPU and turn it into a picture you can see.
- This task is performed by Graphic Card which is built into motherboard.
- A graphics card's job is complex, but its principles and components are easy to understand.
- The CPU, working in conjunction with software applications, sends information about the image to the graphics card.
- The graphics card decides how to use the pixels on the screen to create the image.
- It then sends that information to the monitor through a cable.
- To make a 3-D image, the graphics card first creates a wire frame out of straight lines. Then, it rasterizes the image.
- It also adds lighting, texture and colour.
- The graphics card accomplishes this task using four main components:
 - A motherboard connection for data and power
 - A processor to decide what to do with each pixel on the screen
 - Memory to hold information about each pixel and to temporarily store completed pictures
 - A monitor connection so you can see the final result